OptiCrop: Smart Agricultural Production Optimization Engine

Pre-requisites

Overview
Before starting the OptiCrop project, ensure the required software, libraries, and basic knowledge are available. This setup enables efficient development, model training, visualization, and deployment.

Software & Libraries

Anaconda Navigator: Manage Python environments, packages, and data science applications.

PyCharm IDE: Develop, debug, and execute Python applications.

NumPy: Numerical computing and multidimensional arrays.

Pandas: Data loading, cleaning, preprocessing, and analysis.

Scikit-learn: Machine learning model development and evaluation.

Matplotlib: Data visualization using charts and graphs.

Seaborn: Statistical data visualization.

Flask: Deploy the trained machine learning model as a web application.

Knowledge Requirements

• Python Programming
• Data Preprocessing and Exploratory Data Analysis (EDA)
• Machine Learning Fundamentals
• Data Visualization
• Flask Web Development
• Basic HTML & CSS (optional for UI)
• Git/GitHub (optional)

Hardware Requirements

• Windows, macOS, or Linux
• Minimum 8 GB RAM (16 GB recommended)
• Intel Core i5 or equivalent
• 5 GB free disk space
• Stable Internet connection

Python Installation Commands

pip install numpy pandas scikit-learn matplotlib seaborn flask

conda create -n opticrop python=3.11
conda activate opticrop
conda install numpy pandas matplotlib seaborn scikit-learn flask

Expected Outcome

After completing the setup, the environment will be ready to import agricultural datasets, preprocess data, train crop prediction models, visualize results, deploy the model using Flask, and test predictions through a user-friendly web interface.